

Hardware-Aware Discovery of Clifford-Deformed Quantum Error Correction Codes

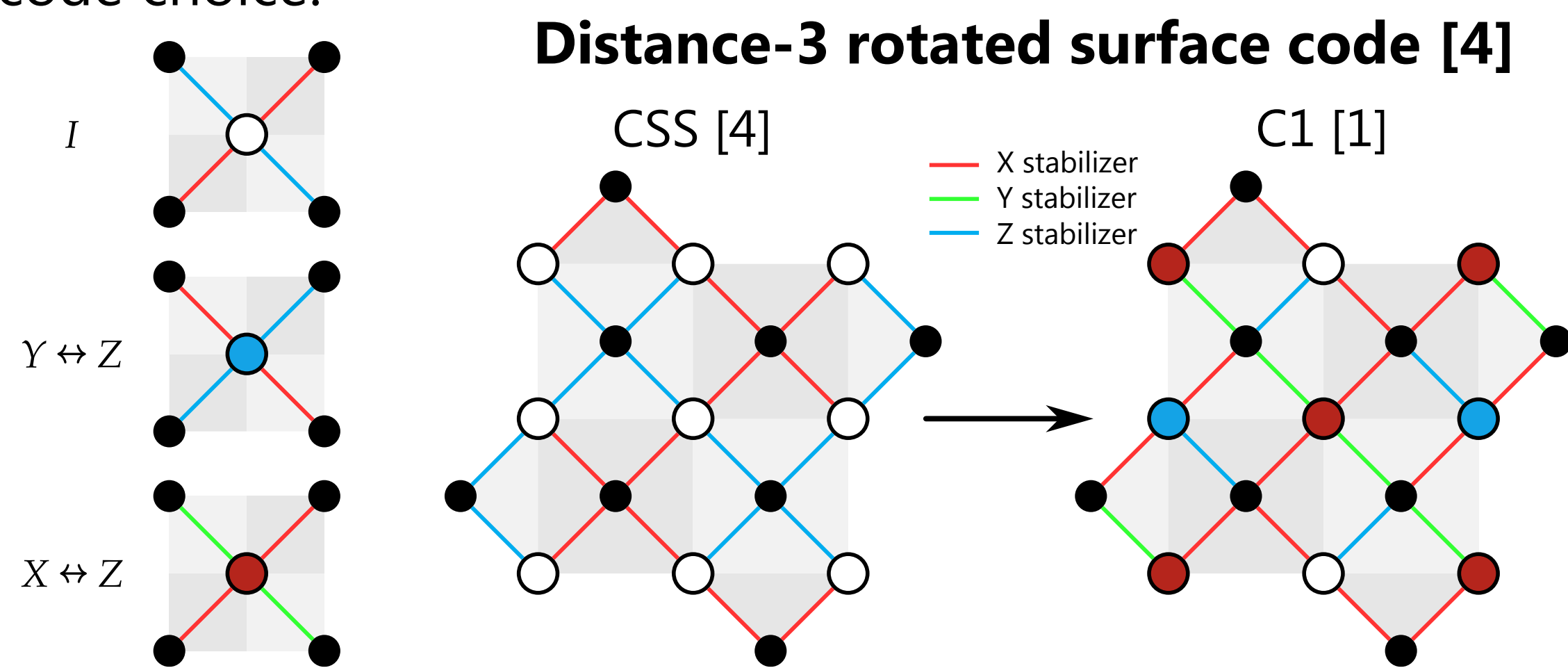
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Background

Clifford-deformations:

- Clifford-deformations (CDs) can reduce Logical Error Rates (LER) under biased noise [1] without increasing the physical qubit count.
- The number of CDs grows as 3^n for an $[[n, k, d]]$ code.
- The Optimal CD depends on the noise model, decoder and QEC code choice.

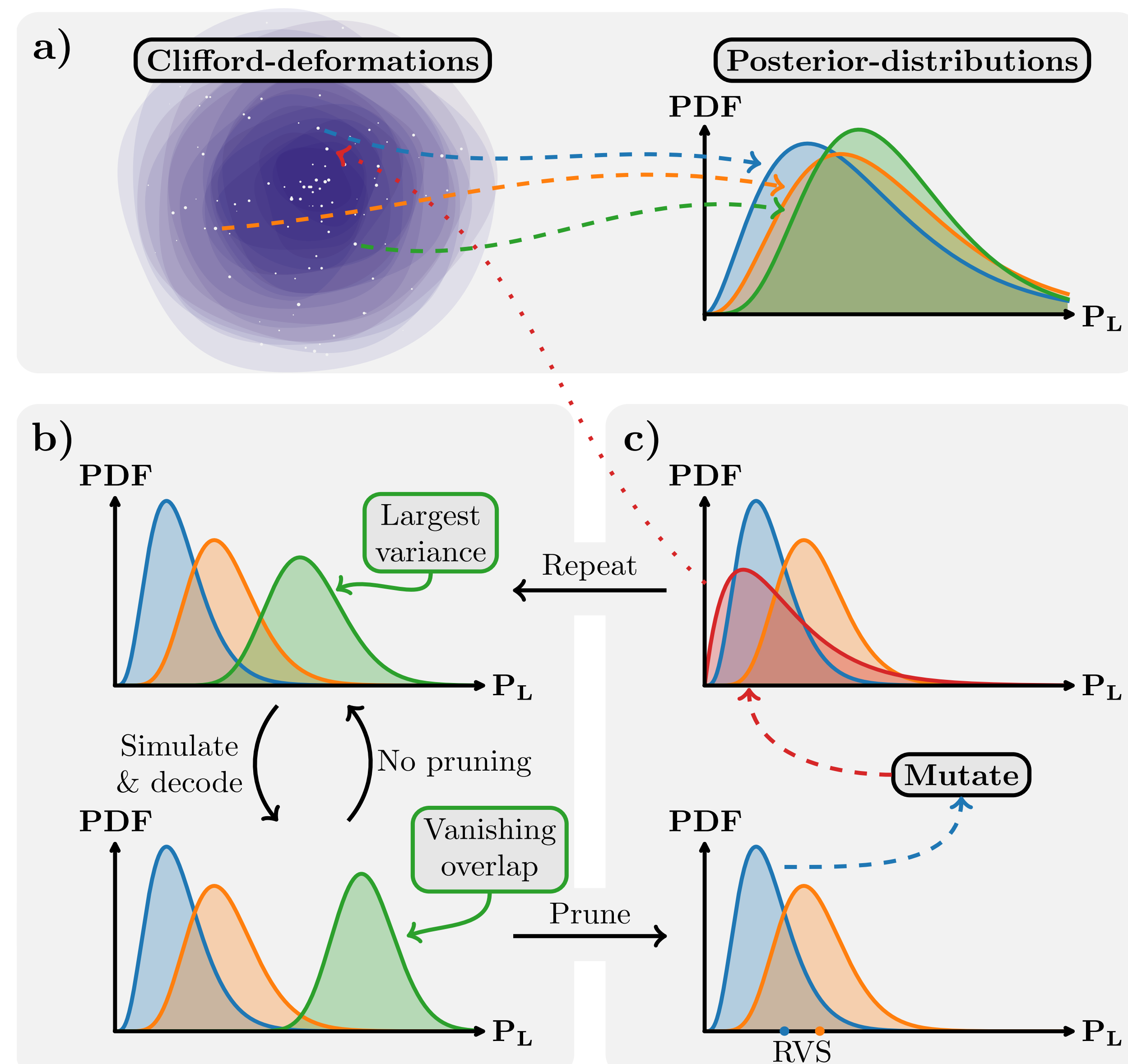


Simulation setup:

- CDs on the distance-3 rotated surface code [1, 4].
- Biased circuit-level STIM [2] noise model.
- Syndrome decoding using PyMatching [3] MWPM-decoder.
- Goal:** Identify CD surface codes with low Logical Error Rates (LER).

Method

Diagram of the Evolution Based Best-arm Identification (EBBI) method.

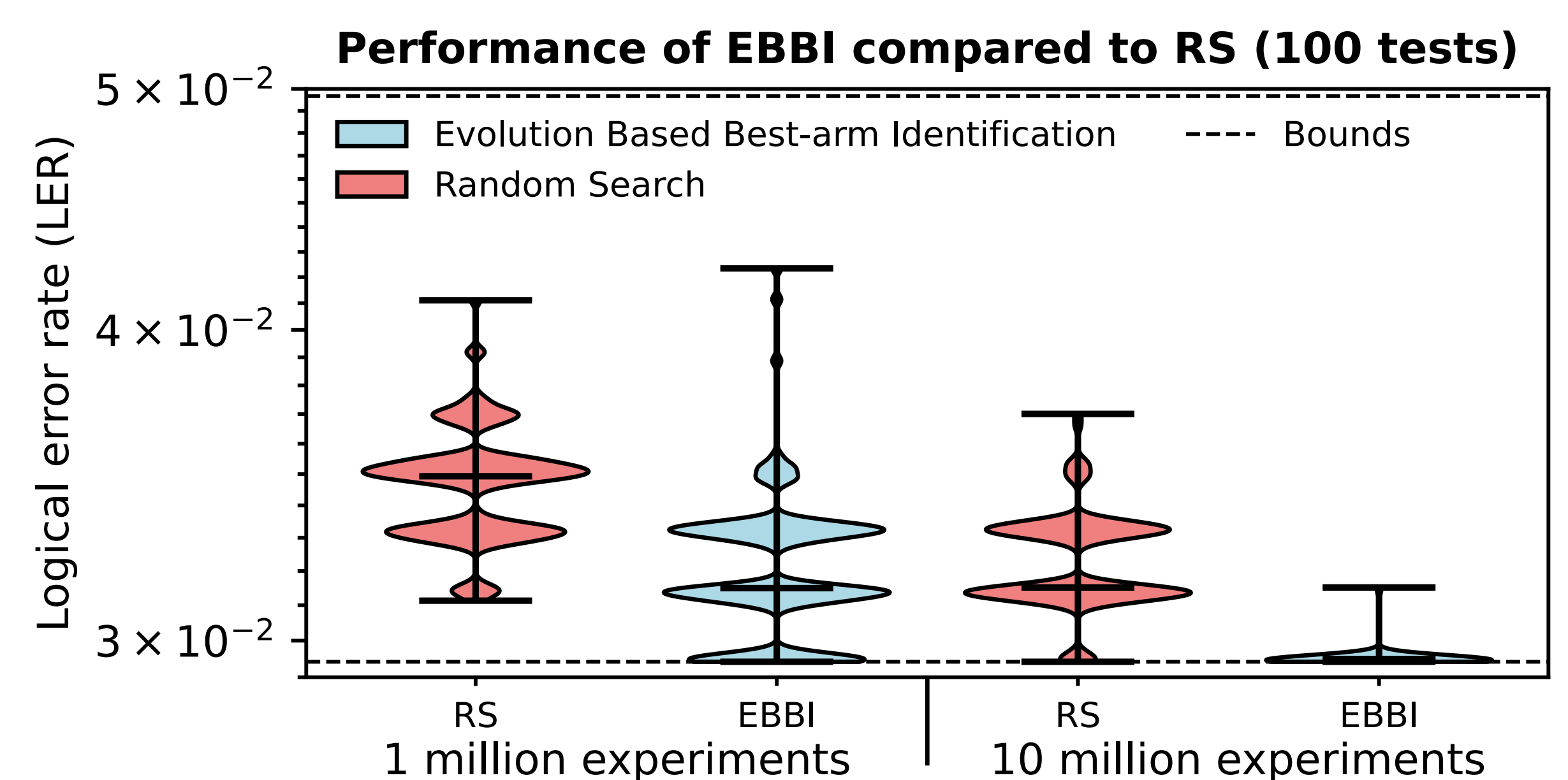


- Initialize:** Select N random Clifford deformations (CDs).
- Sample:** Evaluate the CD with the largest variance.
- Prune:** Remove CDs with low overlap. If none qualify, return to Step 2.
- Mutate:** Replace a pruned CD with a mutation of the best-performing CD. Continue until the budget is exhausted.
- Output:** Return the CD with the lowest expected LER (P_L)

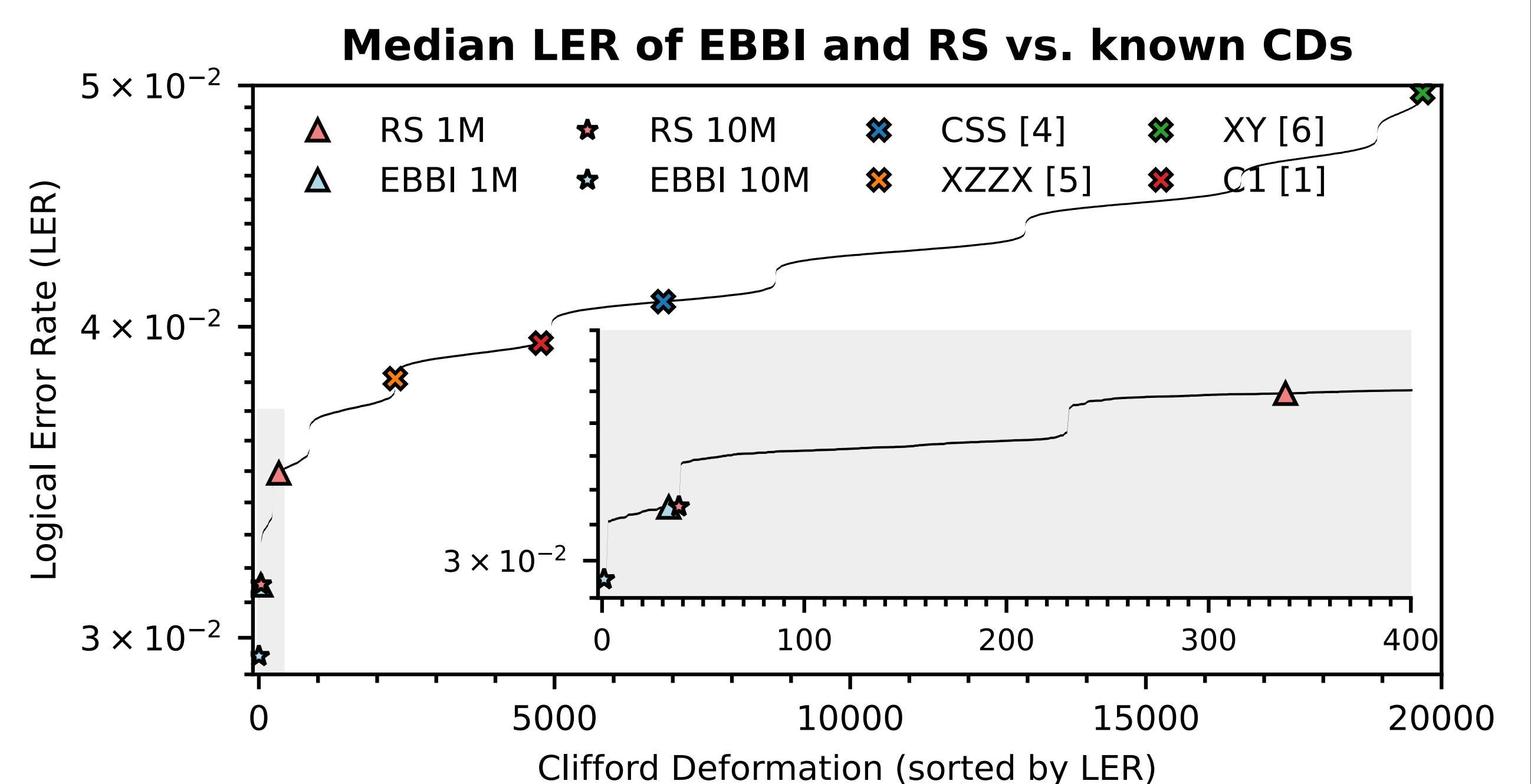
Next Steps

- Expand the evaluation to **larger CD** surface codes.
- Evaluate efficiency of **EBBI** on other codes such as **QLDPC**.
- Explore using **EBBI** directly on **quantum hardware**.

Key results



- Random Search (RS):** Explore N randomly selected Clifford-deformations (CDs) using a fixed, optimized budget per CD.
- Evolution Based Best-arm Identification (EBBI):** Optimized parameters (*number of simultaneous candidates, pruning criteria and mutation rate*) were used for the plotted results.



Takeaway

EBBI identifies low-LER Clifford-deformations using **~10x** fewer experiments than **RS**.

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References:

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- [2] C. Gidney, Quantum 5, 497 (2021)
- [3] O. Higgott and C. Gidney, Quantum 9, 1600 (2025)
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- [6] Tuckett, David K., et al., Physical Review X 9.4, 041031 (2019)